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M3DOCRAG: Multi-modal Retrieval is What You Need for Multi-page Multi-document Understanding

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Abstract

Document visual question answering (DocVQA) pipelines that answer questions from documents have broad applications. Existing methods focus on handling single-page documents with multi-modal language models (MLMs), or rely on text-based retrieval-augmented generation (RAG) that uses text extraction tools such as optical character recognition (OCR). However, there are difficulties in applying these methods in real-world scenarios: (a) questions often require information across different pages or documents, where MLMs cannot handle many long documents; (b) documents often have important information in visual elements such as figures, but text extraction tools ignore them. We introduce M3DOCRAG*, a novel multi-modal RAG framework that flexibly accommodates various document contexts (closed-domain and open-domain), question hops (single-hop and multi-hop), and evidence modalities (text, chart, figure,* etc.*).* M3DOCRAG *finds relevant documents and answers questions using a multi-modal retriever and an MLM, so that it can efficiently handle single or many documents while preserving visual information. Since previous DocVQA datasets ask questions in the context of a specific document, we also present* M3DOCVQA*, a new benchmark for evaluating open-domain DocVQA over 3,000+ PDF documents with 40,000+ pages. In three benchmarks (*M3DOCVQA*/MMLongBench-Doc/MP-DocVQA), empirical results show that* M3DOCRAG *with ColPali and Qwen2-VL 7B achieves superior performance than many strong baselines, including state-of-the-art performance in MP-DocVQA. We provide comprehensive analyses of different indexing, MLMs, and retrieval models. Lastly, we qualitatively show that* M3DOCRAG *can successfully handle various scenarios, such as when relevant information exists across multiple pages and when answer evidence only exists in images.*

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1. Introduction and Background

Document visual question answering (DocVQA) [14, 31, 40, 42, 57] is a multi-modal task that answers textual questions by interpreting information contained within document images. Existing methods on DocVQA either focus on visual question answering (VQA) on a single-page document (Fig. 1 (a)) or extract text from documents (*e.g.*, via optical character recognition (OCR) [43, 53] or PDF text extraction [18, 49]) and use retrieval-augmented generation (RAG) [35], where a retrieval model finds relevant paragraphs and a language model answers questions given the paragraphs (Fig. 1 (b)). However, there are difficulties in applying these methods in real-world document understanding scenarios: (a) questions often require information across different pages or documents, where existing VQA methods cannot handle many long documents; (b) some documents feature complex visual formats such as tables, charts, and mixed layouts, but text extraction methods such as OCR ignore these nuances, leading to incomplete or inaccurate document interpretations. Accurately and efficiently answering questions across numerous, lengthy documents with intricate layouts would greatly benefit many domains such as finance, healthcare, and law, where document AI assistants can streamline the daily processing of large volumes of documents, improving productivity and enabling faster, more informed decision-making.

To overcome these limitations of existing DocVQA approaches, we introduce **M3DOCRAG** (**M**ulti-modal **M**ulti-page **M**ulti-**Doc**ument **R**etrieval-**A**ugmented **G**eneration; Sec. 2), a novel multi-modal RAG framework that flexibly accommodates various document contexts (closed-domain and open-domain), question hops (single-hop and multi-hop), and evidence modalities (text, chart, figure, *etc.*). As illustrated in Fig. 1 (c), the M3DOCRAG framework retrieves relevant document pages using a multi-modal retrieval model, such as ColPali [17], and generates answers to questions from the retrieved pages using a multi-modal language model (MLM), such as Qwen2-VL [59]. M3DOCRAG operates in three stages: In (1) document

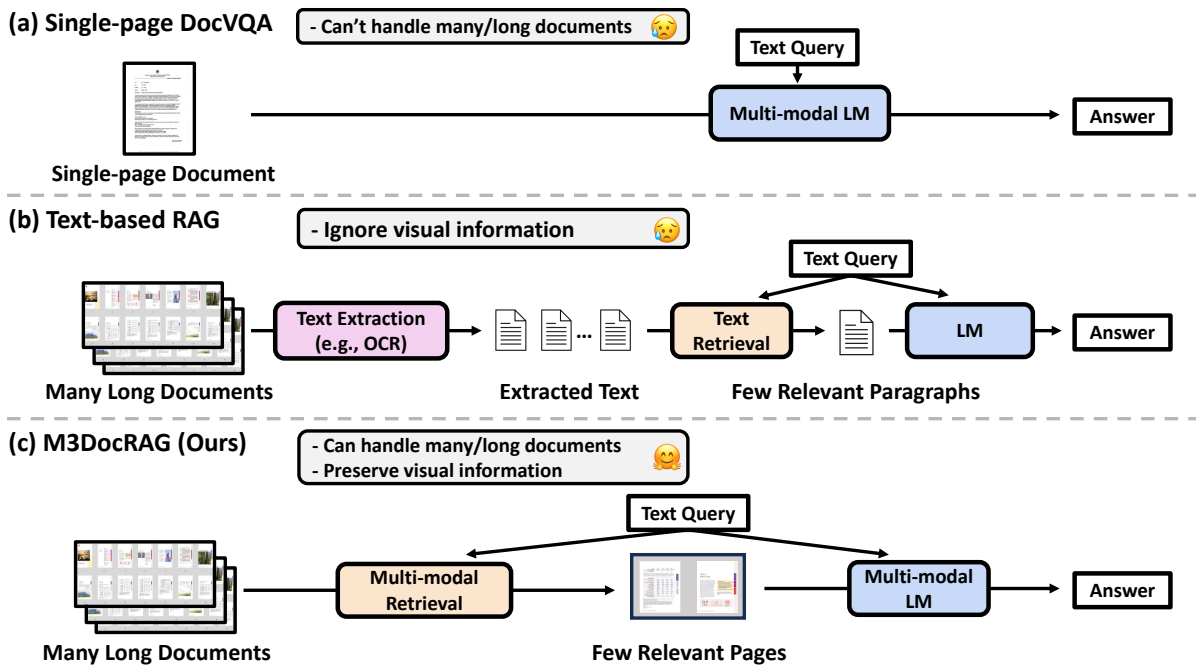

Figure 1. Comparison of multi-modal document understanding pipelines. Previous works focus on **(a) Single-page DocVQA** that cannot handle many long documents or **(b) Text-based RAG** that ignores visual information. Our **(c) M3DocRAG** framework retrieves relevant documents and answers questions using multi-modal retrieval and MLM components, so that it can efficiently handle many long documents while preserving visual information.

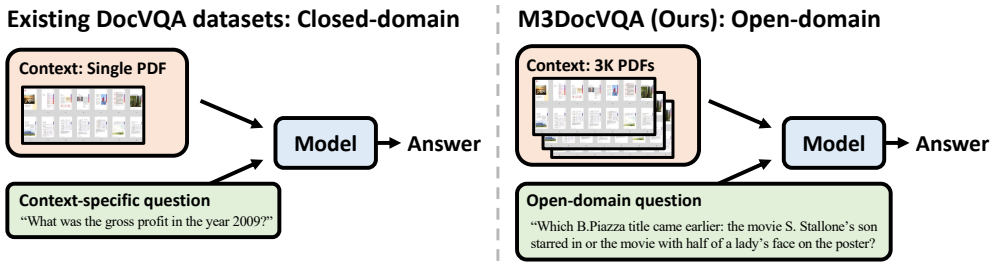

Figure 2. Comparison of existing DocVQA datasets (left; *e.g.*, DocVQA [42]) and our **M3DocVQA** dataset (right). In contrast to previous DocVQA datasets that have questions that are specific to a single provided PDF (*e.g.*, "What was the gross profit in the year 2009?"), M3DocVQA has information-seeking questions that benchmark open-domain question answering capabilities across more than 3,000 PDF documents (*i.e.*, 40,000+ pages).

embedding (Sec. 2.1), we convert all document pages into RGB images and extract visual embeddings (*e.g.*, via ColPali) from the page images. In (2) page retrieval (Sec. 2.2), we retrieve the top-K pages of high similarity with text queries (*e.g.*, MaxSim operator for ColPali). For the open-domain setting, we create approximate page indices, such as inverted file index (IVF) [52, 66], for faster search. In (3) question answering (Sec. 2.3), we conduct visual question answering with MLM to obtain the final answer. Please also see Fig. 3 for the detailed illustration of the framework. M3DocRAG can flexibly handle DocVQA in both closed domain (*i.e.*, a single document) and open-domain (*i.e.*, a large corpus of documents) settings.

While M3DocRAG framework supports DocVQA in an open-domain setting, the existing DocVQA datasets are not adequate for this setting, since their questions are in the context of a specific document, such as "What was the gross profit in the year 2009?" [14, 40, 42, 57], as illustrated in Fig. 2 (left). Hence, we also introduce **M3DocVQA** (**M**ulti-modal **M**ulti-page **M**ulti-**Doc**ument **V**isual **Q**uestion **A**nswering), an open-domain dataset that significantly raises the challenge of DocVQA to answering questions from a large document corpus (Sec. 3). By extending the MultimodalQA dataset's [54] closed-domain context to an open-domain setting, M3DocVQA introduces 2,441 multi-hop questions spanning 3,368 PDF doc-

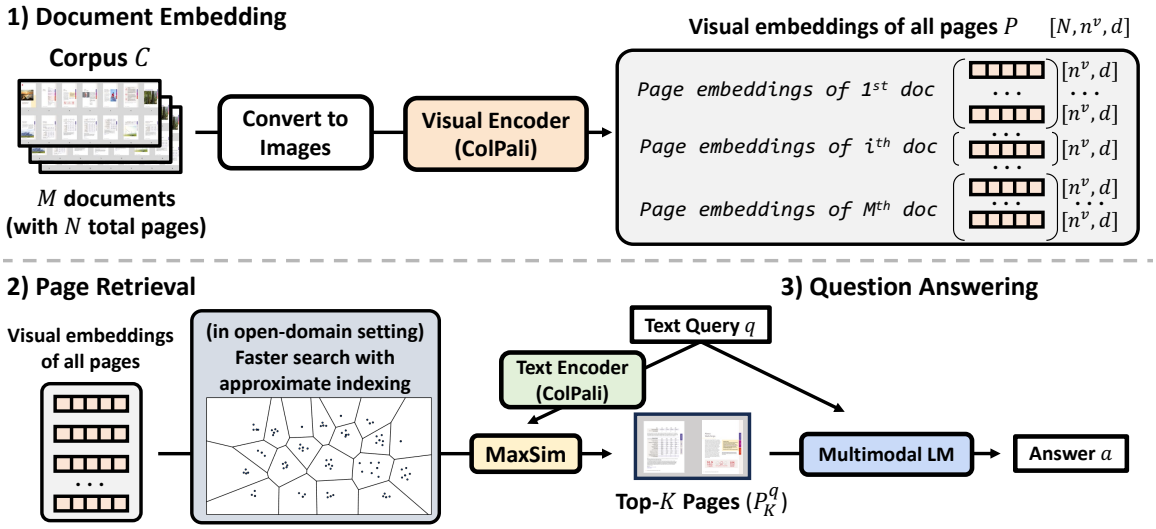

Figure 3. Our M3DocRAG framework (Sec. 2) consists of three stages: (1) document embedding (Sec. 2.1), (2) page retrieval (Sec. 2.2), and (3) question answering (Sec. 2.3). In **(1) document embedding**, we extract visual embedding (with ColPali) to represent each page from all PDF documents. In **(2) page retrieval**, we retrieve the top-K pages of high relevance (MaxSim scores) with text queries. In an open-domain setting, we create approximate page indices for faster search. In **(3) question answering**, we conduct visual question answering with multi-modal LM (*e.g.* Qwen2-VL) to obtain the final answer.

uments, which collectively contain over 41,005 pages of diverse multi-modal content, including text, images, and tables. This dataset presents real-world challenges by requiring models to navigate complex reasoning paths across pages and within various types of document elements, better reflecting the intricacies of document understanding.

To demonstrate the effectiveness of M3DocRAG, we compare M3DocRAG with state-of-the-art baselines in three benchmarks: M3DocVQA, MMLongBench-Doc [40], and MP-DocVQA [57], which cover both open-domain (Sec. 5.1) and closed-domain (Sec. 5.2) DocVQA settings. Experiment results show that M3DocRAG with ColPali and Qwen2-VL 8B achieves superior performance than many strong baselines, including the state-of-the-art performance in MP-DocVQA. We also provide a comprehensive analysis (Sec. 5.3) about different indexing, MLMs, and retrieval components. Finally, we show qualitative examples (Sec. 5.4) where M3DocRAG can successfully handle various scenarios, such as when the relevant information exists across multiple pages and when answer evidence only exists in images. Overall, M3DocRAG is an effective, efficient, and flexible framework for answering questions from multi-modal documents in various settings.

2. M3DocRAG: A Unified Framework for Multi-modal, Multi-page, Multi-document Understanding

We propose **M3DocRAG**, a novel multi-modal RAG framework that flexibly accommodates various document contexts (closed-domain and open-domain), question hops (single-hop and multi-hop), and evidence modalities (text, chart, figure, *etc.*). As illustrated in Fig. 3, M3DocRAG operates in three stages: (1) encoding document images into visual embeddings (Sec. 2.1), (2) retrieving relevant document pages (Sec. 2.2), and (3) generating answers to questions based on the retrieved pages (Sec. 2.3). Below, we explain the problem definition and the details of each stage.

Problem definition. We define a corpus of documents as $C = \{D_1, D_2, \ldots, D_M\}$, where M is the total number of documents, and each document D_i consists of a set of pages, P_i, represented as RGB images. From the documents in C, we construct a global set of page images $P = \bigcup_{i=1}^{M} P_i = \{p_1, p_2, \ldots, p_N\}$, where each p_j represents an individual page image, and N is the total number of page images across all documents in C (*i.e.*, $N = \sum_{i=1}^{M} |P_i|$). The objective of M3DocRAG is to accurately answer a given question q using the multi-modal information available in the corpus of documents C. First, we identify P_K^q, the top K ($\ll N$) pages that are most relevant to answering the query q from the global page set P. Then, we obtain the final answer with a question answering model that takes retrieved page images P_K^q and query q as inputs. The problem of question answering can be categorized into two settings with different document context sizes:

Closed-domain question answering – The query q should be answerable from a given single document D_i. The retrieval model outputs the top K relevant page images

P_K^q, from the page images P_i of the document D_i.

Open-domain question answering – The query q may require information from single or multiple documents within the entire document corpus C. The retrieval model outputs the top K relevant page images P_K^q from the entire set of page images P.

2.1. Document Embedding

In M3DocRAG, both textual query q and page images P are projected into a shared multi-modal embedding space using ColPali [17]. ColPali is a multi-modal retrieval model based on a late interaction mechanism, which encodes the text and image inputs into unified vector representations and retrieves the top K most relevant images. ColPali adopts both training objective and similarity scoring from ColBERT [29, 50], which utilizes a shared architecture to encode either textual or visual inputs. In our framework, each page $p \subseteq P_i$ of a document D_i is treated as a single image with fixed dimensions (width $\times$ height).

From an image of a page, we extract a dense visual embedding $E^p \in \mathbb{R}^{n^v \times d}$, where n^v represents the number of visual tokens per page (which remains constant across all pages), and d denotes the embedding dimension (*e.g.*, 128). For a textual query q, we similarly obtain an embedding $E^q \in \mathbb{R}^{n^q \times d}$, where n^q is the number of text tokens.

For efficiency, we treat each page of a document independently. This allows us to flatten all pages in the document corpus C into a single page-level embedding tensor: $E^{\mathrm{C}} \in \mathbb{R}^{N \times n^v \times d}$, where N represents the total number of pages in the entire document corpus, n^v is the number of visual tokens per page, and d is the embedding dimension. M3DocRAG can flexibly adapt to different retrieval settings, such as a single-page document ($N = 1$), a single document with multiple pages (*e.g.* $N = 100$), and a large corpus of multi-page documents (*e.g.* $N > 1,000$).

2.2. Page Retrieval

The relevance between the query q and the page p is computed using the `MaxSim` score $s(q, p)$:

$$s(q,p) = \sum_{i=1}^{n^q} \max_{j \in [n^v]} E^q_{i,\cdot} \cdot E^p_{j,\cdot}$$

where $\cdot$ denotes the dot product, and $E_{i,\cdot} \in \mathbb{R}^d$ denotes the i-th row (vector) of the embedding matrix $E \in \mathbb{R}^{n \times d}$. We then identify P_K^q, the top K ($\ll N$) pages that are most relevant to answering the query q; *i.e.* we search K pages scoring highest $s(q, p)$. That is,

$$P_K^q = \{p_1^q, p_2^q, \ldots, p_K^q\} = \text{argtop-k}_{p \in P}\, s(q,p)$$

Approximate indexing for open-domain page retrieval. Searching pages over in a large document corpus can be time-consuming and computationally expensive. When a faster search is desired, we create page indices offline by applying approximate nearest neighborhood search, based on Faiss [16, 26]. We use exact search for closed-domain page retrieval and employ inverted file index (IVF) [52, 66] (`IVFFlat` in Faiss) for an open-domain setting, which could reduce page retrieval latency from 20s/query to less than 2s/query when searching across 40K pages. See Sec. 5.3 for a detailed comparison of speed-accuracy tradeoffs across different indexing methods.

2.3. Question Answering

We run visual question answering by giving the text query q and retrieved page images P_K^q to a multi-modal language model to obtain the final answer. For this, we employ multi-modal language models (*e.g.* Qwen2-VL [59]) that consist of a visual encoder $\texttt{Enc}^{\texttt{Vis}}$ and a language model `LM`. The visual encoder takes K-retrieved page images P_K^q as inputs and outputs visual embeddings (different from ColPali encoder's outputs). The language model takes the visual embeddings and text embeddings of query q as inputs and outputs the final answer a in the autoregressive manner:

$$a = \texttt{LM}(\texttt{Enc}^{\texttt{Vis}}(P_K^q), q).$$

3. M3DocVQA: A New Benchmark for Open-domain Document Understanding

We present **M3DocVQA** (**M**ulti-modal **M**ulti-page **M**ulti-**Doc**ument **V**isual **Q**uestion **A**nswering), a new open-domain DocVQA benchmark designed to evaluate the ability to answer questions using multi-modal information from a large corpus of documents.

As illustrated in Fig. 2, existing DocVQA datasets [31, 40, 42, 57] primarily focus on evaluating question answering within the context of a single document (*i.e.*, closed-domain). These datasets are not well-suited for benchmarking open-domain visual question answering, where relevant information, often in multiple modalities such as text, images, and tables, must be retrieved from multiple documents. This limitation stems from their questions being designed around specific content on certain pages within a single document. In real-world scenarios, users often seek answers that span across multiple documents and modalities, making open-domain settings critical. However, the questions in the existing DocVQA datasets are not applicable in such an open-domain setting. For example, a question from MP-DocVQA, such as *"What was the gross profit in the year 2009?"* assumes that the model already has access to specific information within the document.

M3DocVQA challenges models in an open-domain DocVQA setting, where they must navigate a large 'haystack' of multi-modal documents and retrieve relevant

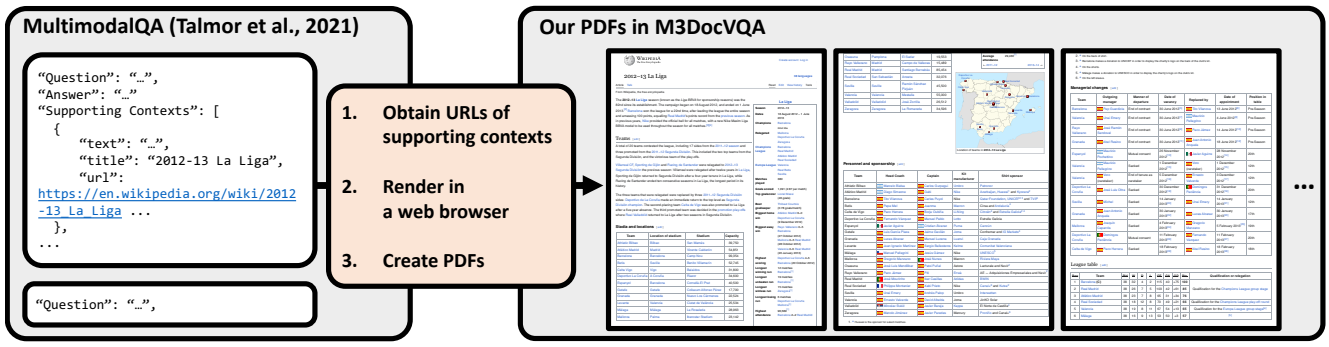

Figure 4. Illustration of PDF collections in M3DocVQA. We first collect the URLs of all supporting contexts (Wikipedia documents) of individual questions of MultimodalQA [54]. Then, we create PDF versions from their URLs by rendering them in a web browser.

information to generate the final answer. The dataset consists of 2,441 multi-hop questions spread across 3,368 PDF documents, totaling 41,005 pages. Each question is supported by evidence found in one or more documents, spanning multiple modalities such as text, images, and tables, capturing the complexity and diversity typical of real-world documents. Additionally, we provide the training split, consisting of 24,162 Wikipedia PDFs. Although the documents in the training split were not utilized in our experiments, they offer future researchers the opportunity to explore even larger-scale retrieval tasks or use the documents for training models, further expanding the potential applications of M3DocVQA.

To create M3DocVQA, we extend the question-answer pairs from a short-context VQA dataset to a more complex setting that includes 1) PDF documents and 2) open-domain contexts. Specifically, we use the question-answer pairs from the development split[1] of MultimodalQA [54], where models answer multi-hop questions based on short multi-modal contexts (*e.g.*, short text passages, 1-2 images, a table) sourced from Wikipedia. We retrieved the URLs of all Wikipedia documents used as context in any of the MultimodalQA development split questions. Then we generated PDF versions of the Wikipedia pages by rendering them in a Chromium web browser [56], using the Playwright Python package [45]. These PDFs retain all vector graphics and metadata, ensuring zoom-in functionality and maintaining operational hyperlinks. In addition, no objects are split between different pages in the resulting PDFs.

While both M3DocVQA and MultimodalQA [54] share the goal of evaluating question answering given multi-modal context, M3DocVQA introduces a more demanding scenario by requiring models to retrieve relevant information from a large set of documents, as opposed to being provided with a short context. In MultimodalQA, models are given short, curated context (*e.g.*, a paragraph from a Wikipedia document) that directly contains the information needed to answer the questions, simplifying the task to reasoning within the provided material. In contrast, M3DocVQA presents an open-domain setting, where models must retrieve information from a diverse collection of 3,368 PDF documents before attempting to answer any question. This not only requires handling large-scale document retrieval but also dealing with multi-modal content–text, images, and tables–distributed across multiple documents. This key distinction highlights M3DocVQA's ability to simulate real-world challenges, where the relevant data is often spread across multiple sources. Consequently, M3DocVQA serves as a robust benchmark for retrieval-augmented generation tasks in document understanding, pushing the boundaries of models to deal with large-scale, multi-modal, and multi-document settings.

[1]The test split of MultimodalQA [54] is unavailable, and previous works have used the development split for comparison.

4. Experiment Setup

Datasets. We benchmark M3DocRAG on three PDF document understanding datasets that represent different scenarios: (1) M3DocVQA (Open-domain DocVQA); (2) MMLongBench-Doc [40] (Closed-domain DocVQA); (3) MP-DocVQA [57] (Closed-domain DocVQA). In M3DocVQA, M3DocRAG processes over 3,000 PDFs, totaling more than 40,000 pages. For MP-DocVQA, models handle a single PDF with up to 20 pages for each question. For MMLongBench-Doc, models handle a single PDF with up to 120 pages for each question.

Evaluation Metrics. For M3DocVQA, we follow the evaluation setup of MultimodalQA [54]. For MMLongBench-Doc [40] and MP-DocVQA [57], we follow their official evaluation setups. For M3DocVQA, we evaluate answer accuracy with exact match (EM) and F1. For MMLongBench-Doc, we extract short answers with GPT4o [46] from the model outputs and report answer accuracy with generalized accuracy (based on a rule-based

Table 1. Open-domain DocVQA evaluation results on M3DocVQA. The scores are based on F1, unless otherwise noted. Index: `FlatIP + IVFFlat`.

Method	# Pages	Evidence Modalities			Question Hops		Overall	
		Image	Table	Text	Single-hop	Multi-hop	EM	F1
Text RAG (w/ ColBERT v2)								
Llama 3.1 8B	1	8.3	15.7	29.6	25.3	12.3	15.4	20.0
Llama 3.1 8B	2	7.7	16.8	31.7	27.4	12.1	15.8	21.2
Llama 3.1 8B	4	7.8	21.0	34.1	29.4	15.2	17.8	23.7
M3DocRAG *(w/ ColPali)*								
Qwen2-VL 7B (Ours)	1	25.1	27.8	39.6	37.2	25.0	27.9	32.3
Qwen2-VL 7B (Ours)	2	**26.8**	**30.4**	**42.1**	41.0	25.2	29.9	34.6
Qwen2-VL 7B (Ours)	4	24.7	**30.4**	41.2	**43.2**	**26.6**	**31.4**	**36.5**

evaluation script covering different answer types) and F1 score. For MP-DocVQA, we report answer accuracy with ANLS [8] and page retrieval with accuracy (same as recall@1, as there is a single page annotation for each question) by submitting the generation results to the test server.[2]

Models. We mainly experiment with the ColPali v1 [17][3] retrieval model and various recent open source multi-modal LMs with <10B parameters, including Idefics 2 [33], Idefics 3 [32], InternVL 2 [12], and Qwen2-VL [59]. We also experiment with a text-based RAG pipeline by combining recent widely used text retrieval and language models: ColBERT v2 [50] and Llama 3.1 [37]. We also compare ColPali v1 with ColQwen v0.1 [17],[4] another recent multi-modal retrieval model that was trained with same objective/dataset as ColPali but initialized with Qwen2-VL 2B [59] backbone. For reproducible evaluation, we use deterministic greedy decoding for answer generation. We compare these multi-modal and text-based RAG pipelines with recent top entries with comparable parameters (<10B) reported on the leaderboards.

Other implementation details. We use PyTorch [47, 48], Transformers [60], and FlashAttention-2 [13] libraries for running models. We use Tesseract [53] for OCR in text RAG baselines, following Ma et al. [40]. We use Faiss [16, 26] for document indexing. We use the pdf2image [6] library to convert each PDF page into an RGB image with a resolution of DPI=144. While all PDF pages in M3DocVQA have the same size – 8.5 (width) × 11 (height) in inches (*i.e.* US letter size) and 1224 (width) × 1584 (height) in pixels, in MP-DocVQA and MMLongBench-Doc datasets, pages have slightly different sizes. To handle this, we resize page images to the most common image size within the dataset – 1700 (width) × 2200 (height) for MP-DocVQA, and to the most common image size within each PDF document for MMLongBench-Doc. All experiments are conducted with a single H100 80GB GPU. We provide up to 4 pages as visual inputs to our multi-modal LMs, the maximum number of images we could fit in the single GPU.

[2] https://rrc.cvc.uab.es/?ch=17&com=tasks
[3] https://huggingface.co/vidore/colpali
[4] https://huggingface.co/vidore/colqwen2-v0.1

5. Results and Key Findings

In the following, we describe experiment results of M3DocRAG and baselines in both open-domain (Sec. 5.1) and closed-domain settings (Sec. 5.2). Next, we provide ablation studies (Sec. 5.3) about different page indexing strategies and different multi-modal LMs and retrieval models. Lastly, we show qualitative examples (Sec. 5.4) where M3DocRAG can tackle M3DocVQA questions whose answer source exists in various modalities.

5.1. Open-domain DocVQA

Multi-modal RAG outperforms text RAG, especially on non-text evidence sources. Table 1 shows the evaluation results on M3DocVQA. As a model needs to find relevant documents from 3,000+ PDFs for each question, we focus solely on RAG pipelines. We observe that our M3DocRAG (ColPali + Qwen2-VL 7B) significantly outperforms text RAG (ColBERT v2 + Llama 3.1 8B), across all different evidence modalities / question hops / # pages. The performance gap is especially big when the evidence involves images, underscoring that M3DocRAG addresses the information loss over non-textual content by text-only pipelines. We also notice that providing more retrieved pages as context generally increases the performance of both text RAG and M3DocRAG (using the top 4 pages gives higher performance than the top 1 and 2 pages).

5.2. Closed-domain DocVQA

Multi-modal RAG boosts long document understanding of MLMs. In MMLongBench-Doc, the models must handle a long PDF document (up to 120 pages) for each ques-

Table 2. Closed-domain DocVQA evaluation results on MMLongBench-Doc. We report the generalized accuracy (ACC) across five evidence source modalities: text (TXT), layout (LAY), chart (CHA), table (TAB), and image (IMG), and three evidence locations: single-page (SIN), cross-page (MUL), and unanswerable (UNA). The scores from non-RAG methods are from Ma et al. [40].

Method	# Pages	Evidence Modalities					Evidence Locations			Overall	
		TXT	LAY	CHA	TAB	IMG	SIN	MUL	UNA	ACC	F1
Text Pipeline											
LMs											
ChatGLM-128k [5]	up to 120	23.4	12.7	9.7	10.2	12.2	18.8	11.5	18.1	16.3	14.9
Mistral-Instruct-v0.2 [25]	up to 120	19.9	13.4	10.2	10.1	11.0	16.9	11.3	24.1	16.4	13.8
Text RAG											
ColBERT v2 + Llama 3.1	1	20.1	14.8	12.7	17.4	7.4	21.8	7.8	**41.3**	21.0	16.1
ColBERT v2 + Llama 3.1	4	23.7	17.7	14.9	**24.0**	11.9	25.7	12.2	38.1	**23.5**	19.7
Multi-modal Pipeline											
Multi-modal LMs											
DeepSeek-VL-Chat [38]	up to 120	7.2	6.5	1.6	5.2	7.6	5.2	7.0	**12.8**	7.4	5.4
Idefics2 [33]	up to 120	9.0	10.6	4.8	4.1	8.7	7.7	7.2	5.0	7.0	6.8
MiniCPM-Llama3-V2.5 [61, 64]	up to 120	11.9	10.8	5.1	5.9	12.2	9.5	9.5	4.5	8.5	8.6
InternLM-XC2-4KHD [15]	up to 120	9.9	14.3	7.7	6.3	13.0	12.6	7.6	9.6	10.3	9.8
mPLUG-DocOwl 1.5 [22]	up to 120	8.2	8.4	2.0	3.4	9.9	7.4	6.4	6.2	6.9	6.3
Qwen-VL-Chat [4]	up to 120	5.5	9.0	5.4	2.2	6.9	5.2	7.1	6.2	6.1	5.4
Monkey-Chat [36]	up to 120	6.8	7.2	3.6	6.7	9.4	6.6	6.2	6.2	6.2	5.6
M3DocRAG											
ColPali + Idefics2 (Ours)	1	10.9	11.1	6.0	7.7	15.7	15.4	7.2	8.1	11.2	11.0
ColPali + Qwen2-VL 7B (Ours)	1	25.7	21.0	18.5	16.4	19.7	30.4	10.6	5.8	18.8	20.1
ColPali + Qwen2-VL 7B (Ours)	4	**30.0**	**23.5**	**18.9**	20.1	**20.8**	**32.4**	**14.8**	5.8	21.0	**22.6**

tion. Since many multi-modal LMs have limited context length, Ma et al. [40] employed a concatenation strategy that combines all screenshot pages into either 1 or 5 images and inputs these concatenated images to multi-modal LMs. Table 2 shows that ColPali + Idefics2 surpass Idefics2 without RAG, as well as all previous multi-modal entries. In addition, ColPali + Qwen2-VL 7B achieves the best scores in overall F1 and most evidence modality/page settings. This demonstrates the effectiveness of multi-modal retrieval over handling many pages by concatenating low-resolution images. As observed in M3DocVQA experiments, we also notice that providing more retrieved pages as context generally increases the performance of both text RAG and M3DocRAG (using the top 4 pages gives higher performance than the top 1 page).

M3DocRAG achieves the state-of-the-art performance in MP-DocVQA. In MP-DocVQA, the models must handle a PDF document of up to 20 pages for each question. Table 3 presents the top-performing entries in the MP-DocVQA test split leaderboard, comparing text-based and multi-modal RAG pipelines. While the text RAG (ColBERT v2 + Llama 3.1) falls short compared to existing approaches, all multi-modal RAG pipelines outperform their text-based counterpart. Notably, the M3DocRAG pipeline (ColPali + Qwen2-VL 7B) delivers the state-of-the-art results on MP-DocVQA. It is interesting that while the existing entries were fine-tuned specifically for MP-DocVQA, the components used in M3DocRAG (ColPali or Qwen2-VL 7B) were not tailored to this dataset – although Qwen2-VL 7B might have been trained on DocVQA [42], which shares some images with MP-DocVQA.

Table 3. Closed-domain DocVQA evaluation results on MP-DocVQA. The RAG methods retrieve a single page to the downstream QA models.

Method	Answer Accuracy ANLS	Page Retrieval R@1
Multimodal LMs		
Arctic-TILT 0.8B [10]	0.8122	50.79
GRAM [9]	0.8032	19.98
GRAM C-Former [9]	0.7812	19.98
ScreenAI 5B [3]	0.7711	77.88
Text RAG		
ColBERT v2 + Llama 3.1 8B	0.5603	75.33
M3DocRAG		
ColPali + Qwen2-VL 7B (Ours)	**0.8444**	**81.05**

5.3. Additional analysis

Different page indexing: speed and accuracy. In Table 4, we analyze the speed and accuracy of ColPali+Qwen2-VL 7B pipeline with different document embedding indexing methods. While the naive indexing with exact search (`FlatIP`) is slow (21s per query), we find that using approximate indexing such as inverted file [52, 66] (`IVFFlat`) and product quantization [27] (`IVFPQ`) can retain most of the accuracy, while making the search significantly faster ($<$ 2s per query). We use

Table 4. Speed-accuracy tradeoff with different indexing strategies on M3DocVQA. Method: ColPali + Qwen2-VL 7B.

# Pages	Indexing	Latency (s) (↓)		Accuracy (↑)	
		Retrieval	VQA	EM	F1
1	FlatIP	21.0	1.1	28.9	33.7
1	FlatIP + IVFFlat	1.8	1.1	27.9	32.3
1	FlatIP + IVFPQ	0.2	1.1	25.9	30.3
2	FlatIP + IVFFlat	1.8	2.4	29.9	34.6
2	FlatIP + IVFPQ	0.2	2.4	29.0	33.5
4	FlatIP + IVFFlat	1.8	4.8	31.4	36.5
4	FlatIP + IVFPQ	0.2	4.8	29.9	34.7

`FlatIP+IVFFlat` indexing by default, and users can choose appropriate indexing methods depending on their deployment requirements.

Table 5. Comparison of different multimodal LMs within M3DocRAG, evaluated across different document understanding benchmarks. For retrieval, we use the top-1 page from ColPali for all datasets. We use `FlatIP+IVFFlat` indexing for M3DocVQA.

Multimodal LMs	M3DocVQA	MMLongBench-Doc	MP-DocVQA
	F1 (↑)	Acc (↑)	ANLS (↑)
Idefics2 8B	27.8	10.8	0.56
Idefics3 8B	31.8	16.4	0.77
InternVL2 8B	30.9	17.3	0.81
Qwen2-VL 7B	**32.3**	**18.8**	**0.84**

Different multi-modal LMs. In Table 5, we compare four different multi-modal LMs in the M3DocRAG framework: Idefics2 8B [33], Idefics3 8B [32], InternVL2 8B [12], and Qwen2-VL 7B [59]. The Qwen2-VL 7B model outperforms other MLMs in all three benchmarks. Thus, we use the model as our default MLM component.

Table 6. Comparison of different multi-modal retrieval models within M3DocRAG framework, evaluated across different document understanding benchmarks. We provide Qwen2-VL 7B with top-4 pages for MMLongBench-Doc/M3DocVQA and top-1 page for MP-DocVQA from the retrieval models. We use `FlatIP+IVFFlat` indexing for M3DocVQA.

Ret. Models	M3DocVQA	MMLongBench-Doc	MP-DocVQA
	F1 (↑)	Acc (↑)	ANLS (↑)
ColPali v1	**36.5**	21.0	0.84
ColQwen v0.1	32.1	**21.5**	**0.86**

Different multi-modal retrieval models. In Table 6, we compare two different multi-modal retrival models in M3DocRAG framework: ColPali v1 and ColQwen v0.1 (see Sec. 4 for details). Both models are trained with the same training objectives but are initialized with different MLM architectures: PaliGemma 2B [7] and Qwen2-VL 2B [59], respectively. We find that ColPali achieves significantly better performance in M3DocVQA, while ColQwen achieves slightly better performance in MP-DocVQA and MMLongBench-Doc. Thus, we use ColPali as our default retrieval model.

5.4. Qualitative Examples

In Fig. 5, Fig. 6, and Fig. 7, we provide qualitative examples of M3DocRAG (ColPali + Qwen2-VL 7B)'s question answering results on several M3DocVQA examples. In Fig. 5, the answer information is only visually stored within the game logo ('man is leaning on a motorcycle'), and M3DocRAG could find the information. In Fig. 6, the question requires multi-hop reasoning across different pages/documents, and M3DocRAG could combine information from multiple retrieved pages. In Fig. 7, although ColPali did not retrieve the page that contains information about a team whose logo features a bat, Qwen-2 VL leverages its own knowledge 'Valencia CF has a logo featuring a bat', and could provide the final answer. Overall, the qualitative examples showcase that M3DocRAG can successfully tackle different questions whose answer sources exist in various modalities.

6. Related Work

Document visual question answering. Mathew et al. [42] proposed document visual question answering (DocVQA) task, where a model extracts information from documents by treating them as images, like in generic visual question answering [1]. Most research on DocVQA focuses on handling a single-page document [22, 23, 30, 34, 41, 42, 55, 58, 63], and it has been now a common practice to include the single-page DocVQA [42] as a part of the image understanding evaluation suite among recent MLMs [7, 12, 20, 32, 46, 59]. Several recent works study applying MLMs for DocVQA on multi-page documents [31, 40, 57]. However, all previous works on DocVQA have focused on handling questions in the context of a specific document, such as "What was the gross profit in the year 2009?" [14, 40, 42, 57]. While this is probably due to the limited context length of the backbone multi-modal LMs, this does not reflect real-world scenarios, where users often ask questions that require information across different pages/documents. We address the limitation and propose M3DocRAG framework and M3DocVQA dataset for effective, efficient, and flexible document understanding under various document contexts (closed-domain and open-domain), question hops (single-hop and multi-hop), and evidence modalities (text, chart, figure, *etc.*).

Question: “SIE Bend Studio's 2019 game cover has man leaning on what?”

ColPali + Qwen2-VL 7B: “motorcycle”

Top 2 pages retrieved by ColPali

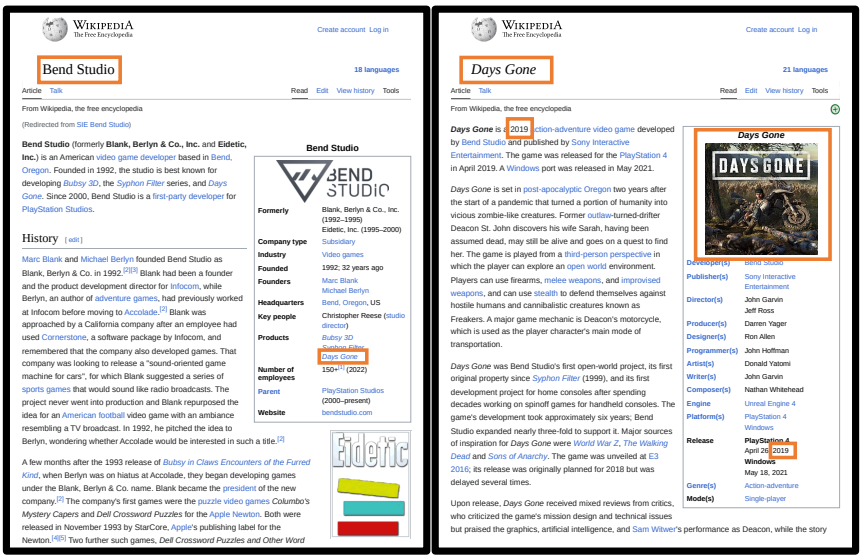

Figure 5. Qualitative example of ColPali + Qwen2-VL 7B on M3DocVQA. Image regions relevant to the question/answer are highlighted with orange boxes. The answer information is only stored visually within the game logo, where a man is leaning on a motorcycle.

Question: “What distance was the AP Warrior fast race at the Del Mar Racetrack?”

ColPali + Qwen2-VL 7B: “Seven Furlongs”

Top 2 pages retrieved by ColPali

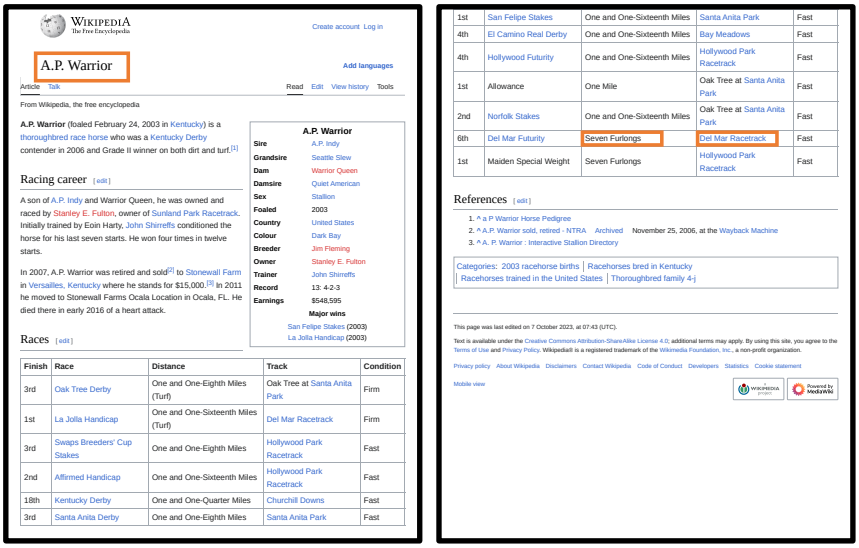

Figure 6. Qualitative example of ColPali + Qwen2-VL 7B on M3DocVQA. Image regions relevant to the question/answer are highlighted with orange boxes. The question requires multi-page/document reasoning.

Figure 7. Qualitative example of ColPali + Qwen2-VL 7B on M3DocVQA. Image regions relevant to the question/answer are highlighted with orange boxes. The VQA component could combine both the retrieved knowledge (Tropi was transferred on 11 July 2017) and its own knowledge (Valencia CF has a logo with a bat) to provide the final answer.

Retrieval-augmented generation. Retrieval-augmented generation (RAG) [35] has emerged as a hybrid approach combining retrieval systems with generative models to improve the quality and relevance of generated content [19]. RAG has been widely studied for open-domain question answering [2, 21, 24, 28, 39, 65], where the community has well-established practices for text-based pipelines. A line of work in VQA studies RAG on visual questions that require world knowledge [11, 44, 51, 62], but their retrieval context is usually generic images and/or short text snippets and does not cover DocVQA settings. To the best of our knowledge, no prior work has explored RAG setting for multi-modal document understanding only with multi-modal models (instead of using OCR methods). Our framework tackles open-domain question answering over documents with complex multi-modal contexts, including textual, tabular, and visual information across different pages and documents.

7. Conclusion

We introduce M3DocRAG, a novel multi-modal RAG framework that flexibly accommodates various document contexts (closed-domain and open-domain), question hops (single-hop and multi-hop), and evidence modalities (text, chart, figure, *etc.*). In M3DocRAG, a multi-modal retrieval model identifies relevant pages from single or multiple documents, which are then processed by a multi-modal language model, where all documents are represented as pixels. Next, we introduce M3DocVQA, the first benchmark that evaluates open-domain multi-modal document understanding capabilities. M3DocVQA consists of 2,000+ questions and 3,000+ PDF documents, and the questions need to be answered with various modalities such as images, text, and tables. Our experiments in three datasets (M3DocVQA, MP-DocVQA, and MMLongBench-Doc) demonstrate significant advantages of M3DocRAG over existing methods, including the state-of-the-art performance in MP-DocVQA. We also provide analysis comparing different indexing strategies, multi-modal LMs, and multi-modal retrieval models. Finally, we show qualitative examples where M3DocRAG can successfully tackle different questions whose answer sources exist in various modalities. We hope that our work encourages future advancements in multi-modal frameworks for document understanding, paving the way for more robust, scalable, and practical solutions in real-world applications.

Ethical Considerations

Limitations. Since our multimodal retrieval models and multimodal LMs were trained with English-heavy datasets, they might not understand prompts or documents written in non-English. While our M3DocRAG framework can benefit many document understanding applications, the model components could present false or biased information. Thus, the framework should be used with human supervision in real-world applications. Note that M3DocRAG is designed with flexibility so that users can update or replace components as more accurate solutions for each element of the framework become available in the future.

Data collection. We do not involve human subjects during data collection. We do not claim ownership/rights of the Wikipedia documents, and we attribute the source Wikipedia document URLs to all pages.

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